Model 2: Price forecast

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What it predicts and how it's scored

Model 2 forecasts prices. It has two outputs, and the second matters more than the first.

1. **The submission column `price_forecast`:** the benchmark price (the average spot price across the material's 3 suppliers) for each week. Worth 12% of the total score.
2. **Landed cost forecasts for each supplier and week,** which the ordering model uses to decide who to buy from and when. These indirectly drive much of the 40% financial score.

For the private leaderboard we forecast weeks 222 to 234 from week 221, so 1 to 13 weeks ahead.

Scoring (MAPE skill). MAPE is the mean absolute percentage error: the average of |forecast − actual| ÷ actual. It is compared with carrying the week-221 benchmark price forward unchanged:

```
\text{price skill} = 1 - \frac{\text{our MAPE}}{\text{MAPE of carry-forward}}
```

Carry-forward scores 0 by definition, and it's a tough baseline over short horizons. So the goal is a simple model that reliably beats it, then moving on: the bigger points are in Models 1 and 3.

Baseline model: follow the futures

The simplest model that can earn points moves each supplier's price from today's spot toward its 4-week futures price, then holds there. `futures_4w` is the market's own price for delivery in 4 weeks, so it's a free forecast.

```
\text{forecast}(h) = \text{spot}_{\text{now}} \times
\left(\frac{\text{futures\_4w}}
{\text{spot}_{\text{now}}}\right)^{\min(h,\,4)/4}
```

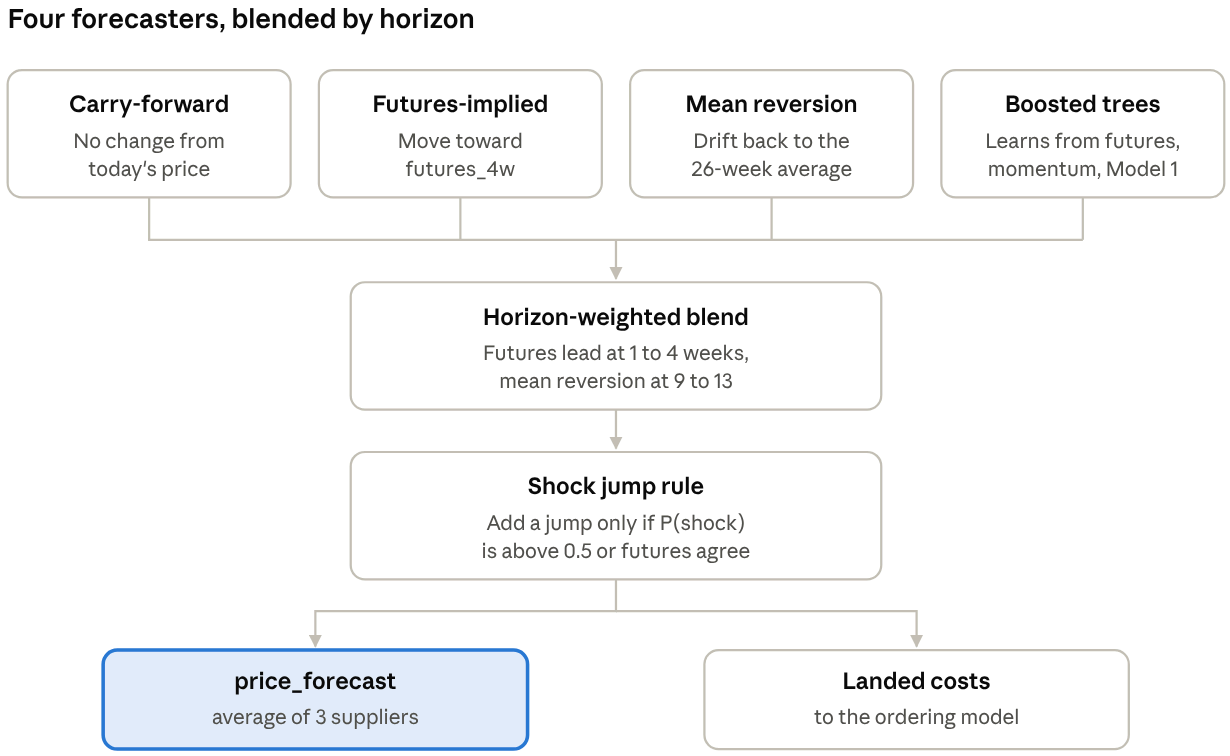
1. For each of the material's 3 suppliers, apply the formula for h = 1 to 13.
2. Average the 3 forecasts to get the benchmark price for each week.

3. Use the same per-supplier forecasts to build landed costs (see the landed cost section).

It's about 10 lines of code. Before trusting it, check in the backtest that futures at week T actually beat spot at week T at predicting spot at week T + 4. If they don't, fall back to carry-forward.

Optimal model: four forecasters, blended by horizon

The full version runs four simple forecasters side by side and weights them differently depending on how far ahead we're looking. The baseline is the futures-implied box on its own.



optimal price model · 4 forecasters, 1 blend

- **Each forecaster predicts the percent change** from the last known spot price, for each supplier separately.
- **Blend weights are chosen separately** for 1 to 4, 5 to 8 and 9 to 13 weeks ahead, based on which members scored best in the backtest. When the evidence is weak, shrink toward carry-forward.

- **Boosted trees go in last,** trained with absolute-error loss, and only stay if they beat the blend without them.

The shock jump rule

Add a price jump for a coming shock only when the shock is more likely than not (probability above 0.5), or when futures already show the market pricing it in.

Why. MAPE rewards being close to the *middle* outcome (the median), not the average. Suppose Model 1 says there's a 30% chance of a shock that raises prices 40%:

- Adding the expected jump (0.3 × 40% = 12%) sounds sensible.
- But in the 70% of cases with no shock, that 12% is pure error.
- Forecasting no jump is wrong only in the 30% of cases where the shock comes, so it scores better on average.

The ordering model handles shock risk separately through its scenarios, so leaving the jump out of this column doesn't leave the plan unprotected.

Landed cost forecasts for the ordering model

Landed cost is what Halcyon actually pays per unit: the spot price plus tariff plus shipping, charged at the week the order is placed. The ordering model needs it for each supplier and week, and getting tariff timing right is what makes it buy before a hike.

1. **Check the formula** in supply_history. Fit:

   ```
   \text{landed\_cost} \approx \text{spot} \times (1 + \text{tariff}) +
   \text{shipping term}
   ```

   where the shipping term is related to that lane's `container_rate`. Look at how well it fits; there may be extra components.
2. **Forecast it** for each supplier and week: the spot forecast from this model, times (1 + the tariff in force that week), plus the shipping term with the container rate carried forward.
3. **Switch tariffs on schedule.** From `announced_effective_week` onward, use `announced_rate` instead of the current rate. This part is bookkeeping, not modeling.
4. **In shock scenarios,** the ordering model multiplies the cost of shocked suppliers by a cost multiplier on top of this forecast.

Testing, and the big shock

Use the same rolling backtest as Model 1, so the models are compared on the same weeks.

- **Origins:** forecast from weeks 156, 169, 182 and 195 (add 208 after Phase 2), each 1 to 13 weeks ahead.
- **Report** MAPE skill overall, per material, and per horizon bucket (1 to 4, 5 to 8, 9 to 13 weeks).
- **Keep a member only if it helps.** If nothing beats futures-implied, stop there.

The big shock. A large shock probably spikes prices, which breaks carry-forward. The best early signal is the futures premium in weeks 209 to 221: if futures for some materials pull well above spot in the Phase 2 data, the market is pricing something in, and the forecast should follow. If futures stay flat, stay close to carry-forward.

Build order

Get the baseline and landed costs done early, then move on: the remaining gains in this column are small next to Models 1 and 3.

- [ ] v0: carry-forward, a valid column right away (5 min)
- [ ] v1: futures-adjusted baseline, likely the biggest single gain (30 min)
- [ ] Landed cost formula check and forecast, needed by the ordering model (1 hr)
- [ ] v2: add carry-forward and mean reversion, fit horizon weights (1 hr)
- [ ] Shock jump rule, once Model 1 produces probabilities (15 min)
- [ ] v3: boosted trees, only if time allows and they beat v2 (1 to 2 hr)